Pro Football Sports Science Research Update

December 2024

Career duration and later-life health conditions among former professional American-style football players. Occupational and Environmental Medicine, Ahead of Print: 1-9, 2024.

Case Report: A time-course observation of reductions in lean body mass following surgery in Elite Rugby Union. Sport Performance and Science Reports, Oct: 1-6, 2024.

Cervical and lumbar disk replacement in athletes: Is it safe to return to play? A systematic review of the scientific literature and lay press. Neurosurgery, 94(1): 4-13, 2024.

Chronotype, sleep hygiene, and sleep characteristics among athletes of different disciplines. International Journal of Sports Medicine, Ahead of Print: 1-11, 2024.

Effects of horizontally versus vertically loaded resistance exercises on performance and muscle architecture. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-8, 2024.

Editorial: Elusive 'usual' warm-up in injury prevention trials: The problem of comparing apples to beans. British Journal of Sports Medicine, Ahead of Print: 1-3, 2024.

Field surface type and season-ending lower extremity injury in NFL players. Translational Sports Medicine, Article ID: 6832213, 2024.

Flywheel resistance training promotes unique muscle architectural and performance-related adaptations in young adults. European Journal of Sport Medicine, Ahead of Print: 1-14, 2024.

From Tissue to System: What constitutes an appropriate response to loading? Sports Medicine, Ahead of Print: 1-19, 2024.

Hamstring Injuries, From the Clinic to the Field: A narrative review discussing exercise transfer. International Journal of Sports Physiology and Performance, Ahead of Print: 1-9, 2024.

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NFL Quarterback Development – The role of competing quarterbacks, rookie playing time, and team quality. The American Economist, Ahead of Print: 1-16, 2024.

Optimizing acute static stretching protocols: Balancing range of motion and strength performance. Current Sports Medicine Reports, 23(11): 363-365, 2024.

Shoulder injury patterns and trends in the National Football League: An epidemiological analysis. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-8, 2024.

Single nucleotide polymorphisms and tendon/ligament injuries in athletes: A systematic review and meta-analysis. International Journal of Sports Medicine, Ahead of Print: 1-19, 2024.

Spatiotemporal kinematics during top speed sprinting in male intercollegiate track and field and team sport athletes. Sports Biomechanics, Ahead of Print: 1-15, 2024.

Sport-related injury and performance enhancing substance use by young athletes. Sports Health, Ahead of Print: 1-15, 2024.

Summary of the 2024 Professionals in Nutrition for Exercise and Sport "10 Questions/10 Experts" Session – Hot Topics for the Paris Olympic Games. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-8, 2024.

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The impact of daylight exposure on injured athletes: Implications for rehabilitation. International Journal of Strength and Conditioning, 4(1): 1-15, 2024.

The increasing medicolegal cost in the care of high-level athletes. Journal of Shoulder and Elbow Surgery. Ahead of Print: 1-18, 2024.

The injury mechanism of Achilles tendon rupture in professional athletes – A video analysis study. Journal of the American Podiatric Medical Association, 114(3): 1-7, 2024.

To challenge or not to challenge? Significance, 21(1): 14-16, 2024.

To jog or to stride: A comparison between two standardized running tests to monitor neuromuscular status in team sport athletes. Journal of Sports Sciences, Ahead of Print: 1-8, 2024.

Towards Achilles Tendon Injury Prevention in Athletes with Structural MRI Biomarkers: A Machine Learning Approach. Sports Medicine – Open, 10: 118, 2024.

Validity of four consumer-grade optical heart rate sensors for assessing volume and intensity distribution of physical activity. Scandinavian Journal of Medicine & Science in Sports, 34: e14756, 2024.

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Career duration and later-life health conditions among former professional American-style football players. Occupational and Environmental Medicine, Ahead of Print: 1-9, 2024.

Objectives: Career duration is often used as a metric of neurotrauma exposure in studies of elite athletes. However, as a proxy metric, career length may not accurately represent causal factors, and associations with health outcomes may be susceptible to selection effects. To date, relationships between professional American-style football (ASF) career length and post-career health remain incompletely characterized.

Methods: We conducted a survey-based cross-sectional cohort study of former professional ASF players. Flexible regression methods measured associations between self-reported career duration and four self-reported health conditions: pain, arthritis, mood and cognitive symptoms. We also measured associations between career duration and four self-reported ASF exposures: prior concussion signs and symptoms (CSS), performance enhancing drugs, intra-career surgeries and average snaps per game. Models were adjusted for age and race.

Results: Among 4189 former players (52±14 years of age, 39% black, 34% lineman position), the average career length was 6.7±3.9 professional seasons (range=1-20+). We observed inverted U-shaped relationships between career duration and outcomes (all $p<0.001$), indicating that adverse health effects were more common among men with intermediate career durations than those with shorter or longer careers. Similar findings were observed for play-related exposures (eg, CSS and snaps).

Conclusions: Relationships between ASF career duration and subsequent health status are non-linear. Attenuation of the associations among longer career players may reflect selection effects and suggest career length may serve as a poor proxy for true causal factors. Findings highlight the need for cautious use of career duration as a proxy exposure metric in studies of former athletes.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Career duration is often used as a proxy metric of neurotrauma exposure in health studies of elite and professional athletes, likely due to ease of recall and verification using public records. However, important considerations related to its use in this setting have not been investigated.

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WHAT THIS STUDY ADDS

⇒ In this study of former professional American style football (ASF) players, former players with both the shortest and longest careers had comparatively lower risk of adverse health outcomes when compared with players with midrange career lengths. In addition, former professional ASF players with the longest careers reported fewer adverse play-related exposures (eg, concussion history and average number of snaps per game).

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Relationships between ASF career duration and subsequent health status may be nonlinear. Specifically, attenuation of associations among longer career players could be related to selection effects resulting in reduced susceptibility of longer career players to certain exposures or to career length serving as a poor proxy for true causal factors.

Case Report: A time-course observation of reductions in lean body mass following surgery in Elite Rugby Union. Sport Performance and Science Reports, Oct: 1-6, 2024.

Aim: We present a case study that highlights the short- and long-term changes in body composition during the rehabilitation from a long-term injury in the presence of an acute nutritional intervention with the secondary aim of understanding the feasibility and value of adopting an injury specific nutritional intervention and DXA timeline for future long-term injuries.

Key take home messages

• Despite prescribing a relatively high caloric intake to overcome energy deficits, and increased protein intake to offset losses of lean body mass, we observed losses of lean body mass, substantially in the injured limb, with increases in fat-mass also observed. MDTs may be more aggressive in their prescription of caloric intakes based on our observations; however, these may need to be tapered given the logistical and consumption issues associated with higher caloric intakes in any sporting environment.

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• Post any surgical intervention within any sporting domain, it is important that a concerted effort is made across all departments within an MDT to minimize the loss of lean body mass during the short- and long-term periods across the recovery process, with an interwoven pathway of agreement across coaching, medical, strength and conditioning, and nutritional stakeholders required to maximize return to function, minimize time-loss and maximally aid a players return to performance.

Cervical and lumbar disk replacement in athletes: Is it safe to return to play? A systematic review of the scientific literature and lay press. Neurosurgery, 94(1): 4-13, 2024.

Background and objectives: Cervical/lumbar total disk replacements (TDRs) are often performed for degenerative conditions but rarely in athletes. Therefore, we sought to conduct a systematic review of athletes undergoing TDRs of both the scientific literature and lay press, with an emphasis on contact sport athletes.

Methods: In accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines, PubMed/Embase/Cochrane/Scopus/Web of Science databases were queried for all primary literature reporting TDRs in athletes, including both cervical/lumbar disk replacements (CDRs/LDRs). Sports were dichotomized into contact vs limited/noncontact. Because of the public nature of injuries in elite athletes, the lay press was also queried.

Results: A total of 488 scientific studies were screened, of which 10 met inclusion criteria. Cervical: seven studies reported CDRs in 53 athletes, of which 7 were professional, 22 semiprofessional, and 24 recreational. Of the seven professional athletes, there was one contact sport athlete (kickboxer). All 7/7 professional and 21/22 semiprofessional athletes successfully returned-to-play at 8-52 weeks without complication. Lumbar: Three studies discussed LDRs in 51 athletes, of which 17 were professional, 6 semiprofessional, and 28 recreational. Of the 17 professional athletes, eight played contact sports (2 boxing, 2 alpine skiing, 2 soccer, judo, rugby). All 17 professional and 6 semiprofessional athletes successfully returned to play at 9-21 weeks. Lay Press: five professional contact sport athletes underwent CDRs, and all returned to play: 3 hockey, 1 mixed-martial arts, and 1 Australian-rules football.

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Conclusion: The scientific literature and lay press revealed 14 professional contact sport athletes who underwent TDR-6 CDRs and 8 LDRs-all with successful return to play. From the little data that exist, it seems that TDR may be safe in elite athletes; however, the small number of patients highlights the major paucity of data on the safety of TDR in elite contact sport athletes.

Chronotype, sleep hygiene, and sleep characteristics among athletes of different disciplines. International Journal of Sports Medicine, Ahead of Print: 1-11, 2024.

Abstract - The objective of the study was to assess chronotype, sleep hygiene, and sleep characteristics of Italian athletes in order to identify differences among sport disciplines (team- vs individual-sports), level of competition (Elite vs Non-Elite) and sexes. Moreover, correlations between chronotype, sleep hygiene and sleep quality were evaluated. Two hundred eighty Italian athletes (Females:113; Males:167) participated in the study. They completed the Pittsburgh Sleep Quality Index (PSQI), the Sleep Hygiene Index (SHI) and the MorningnessEveningness Questionnaire (MEQ). Among the athletes, 66.4% were intermediate-type, 21.8% were morning-type and 11.8% were evening-type. 55.4% of the sample was classified as good sleeper (PSQI<5), while 16.1% had poor sleep hygiene (SHI≥35). Females had higher total sleep time (p=0.022) than males. Individual-sports showed more morning-oriented score (p=0.001) and lower PSQI (p=0.006) and SHI (p<0.001) scores, with earlier timing of sleep (bedtime:p<0.001; wake-up time:p<0.001) compared to teamsports. Elite athletes exhibited greater sleep efficiency (p=0.009), and lower sleep latency (p=0.018) and wake after sleep onset (p=0.043) than Non-Elite athletes. Additionally, significant correlations were found between PSQI and SHI (r=0.43, p<0.001), and between MEQ and SHI (r=-0.35, p<0.001). This study yields crucial insights into the sleep behaviors of Italian athletes, revealing that nearly half of the sample were bad sleepers.

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Effects of horizontally versus vertically loaded resistance exercises on performance and muscle architecture. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-8, 2024.

Background: The aim of this study was to compare a training program based on horizontally (HT) versus vertically (VT) loaded exercises on performance and muscle architecture of the lower body muscles.

Methods: Nineteen resistance trained individuals were randomly assigned to HT (N.=10; age: 25.9±4.2 y; body mass: 72.7±11.4 kg; height: 174.0±6.0 cm) or VT group (N.=9; age: 26.9±4.4 y; body mass: 76.2±10.8 kg; height: 174.2.0±5.8 cm). Both 6-week training programs included 4 training sessions per week and were equated for the total number of repetitions. One repetition maximum (1RM) was assessed for squat and hip thrust, together with vertical and horizontal jumps and sprint. Muscle thickness (MT) and echo intensity (EI) of vastus lateralis, vastus medialis and gluteus were also evaluated pre- and post-training period.

Results: A significantly greater increase in 1RM hip thrust was detected in HT (+17.9%; P=0.004) while greater increases in 1RM squat were found in VT (+10.5%; P=0.007). A greater increase (P=0.009) in vastus medialis MT was detected in VT (4.1%) compared to HT (-7.9%). Similar increases in MT of gluteus were registered in both groups (P<0.05). A greater improvement in standing long jump (P=0.004) was detected in HT (+7.6%) compared to VT (+1.6%), while both groups significantly improved vertical jump performance. Combining both groups, strong correlations were detected between gluteus EI and 20-m sprint (r=0.79; P<0.001).

Conclusions: Results indicate that HT was more effective than VT for horizontal jumps while both HT and VT were equally effective on vertical jumps. Both HT and VT promoted similar changes in muscle architecture of the gluteus, but not of the vastus medialis.

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Editorial: Elusive 'usual' warm-up in injury prevention trials: The problem of comparing apples to beans. British Journal of Sports Medicine, Ahead of Print: 1-3, 2024.

THE ELUSIVE 'USUAL' WARM-UP - Comparing apples to oranges is rational when comparing different fruits; comparing apples to beans is something else entirely. Likewise, injury prevention trials proposing warm-up interventions should strive to deliver reasonable comparisons. The contents and dosage of the intervention and control warm-ups may vary, especially in the context of pragmatic trials, but should be appropriately described to afford transparency and comparability. So, what is this elusive entity known as the 'usual' warm-up in injury prevention trials? And how may this affect the interpretation of the findings? This editorial explores the context and importance of the 'usual' warm-up in injury prevention trials and provides recommendations to harmonize future research.

Field surface type and season-ending lower extremity injury in NFL players. Translational Sports Medicine, Article ID: 6832213, 2024.

Abstract - There is growing concern over the safety of artificial turf when it comes to the incidence of player injuries. The artificial surfaces can withstand more play, are cheaper to maintain, and are more predictable. However, there is concern that this beneficial durability comes at the expense of the forgiveness of the surface, leading to more injuries. In this study, we aim to compare the incidence of in-game season-ending lower extremity injuries on natural and artificial playing surfaces in the National Football League (NFL) during the 2020, 2021, and 2022 seasons. For this study, we used publicly available data to determine and classify the specific injury, where the injury occurred, field surface type, and duration of recovery. All data were collected, and significance was determined using two-sample T-tests. Only in-game injuries were included in this study. Over the course of 2020, 2021, and 2022, there were 199 season-ending lower-extremity injuries. Of these, 79 occurred on natural turf (39.7%) and 120 on artificial turf (60.3%). Of the 891 games played in this three-year period, 396 were played on natural turf (44.4%) and 495 were played on artificial turf (55.6%). Natural turf saw an injury rate of 0.199 per game, and artificial turf saw 0.242 injuries per game. We determined that there is no significant difference in rates of season-ending lower-extremity injury between artificial and natural turf. Any

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perceived difference in injury rates could possibly be attributed to the increased amount of play on artificial surfaces and anecdotal evidence related to high profile cases. While there is no significant difference in incidence, surfaces should continue to be monitored and regulated for traits such as hardness, and player preferences should be considered for qualities that are not quantifiable.

Flywheel resistance training promotes unique muscle architectural and performance-related adaptations in young adults. European Journal of Sport Medicine, Ahead of Print: 1-14, 2024.

Abstract - The purpose of this study was to examine the skeletal muscle hypertrophic, architectural, and performance-related adaptations in response to volume-matched, total-body flywheel versus traditional resistance training in a randomized, non-exercise controlled study in physically active young adults. Thirty-one healthy young adults (24 ± 3 y) were randomized to 10 weeks of traditional resistance training (TRT; n = 7F/5M), flywheel training (FWRT; n = 7F/4M), or a habitual activity control (CON; n = 5F/3M). Maximal voluntary isometric torque (MVIT), one repetition-maximum (1RM) for the free weight squat and bench press, three repetition work maximum (3Wmax) for the flywheel squat and bench press, countermovement jump height, and broad jump distance, as well as site-specific muscle hypertrophy, fascicle length (FL), and pennation angle, were measured. Both TRT and FWRT increased MVIT ($p \leq 0.021$) and FFM ($p \leq 0.032$) compared to CON. However, TRT promoted superior improvements in free weight squat and bench 1RM ($p < 0.001$), and FWRT improved flywheel 3Wmax squat and bench ($p < 0.001$). FWRT increased the FL and cross-sectional area of the distal VL, countermovement jump height, and broad jump distance ($p \leq 0.048$), whereas TRT increased the pennation angle and cross-sectional area of the proximal VL. Therefore, 10 weeks of volume-matched, total-body traditional, and flywheel resistance training similarly increased maximal isometric strength and fat-free mass. However, FWRT promoted unique skeletal muscle architectural adaptations that likely contributed to region-specific VL hypertrophy and jump performance improvements. Thus, FWRT provides a novel training stimulus that promotes architectural adaptations that support improved athletic performance in a manner that is not provided by traditional resistance exercise training.

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Highlights

- Participants using the flywheel isoinertial training device developed unique skeletal muscle adaptations in the vastus lateralis compared to the traditional resistance training group.
- The flywheel group also increased both their countermovement jump height and broad jump distance, whereas there were no improvements in the traditional resistance training group.
- The traditional training group had greater improvements in the free weight back squat and bench press compared to the flywheel group.
- Both groups improved fat-free mass and isometric strength to a similar degree compared to the habitually exercising control group.

From Tissue to System: What constitutes an appropriate response to loading? Sports Medicine, Ahead of Print: 1-19, 2024.

Abstract - Optimal loading involves the prescription of an exercise stimulus that promotes positive tissue adaptation, restoring function in patients undergoing rehabilitation and improving performance in healthy athletes. Implicit in optimal loading is the need to monitor the response to load, but what constitutes a normal response to loading? And does it differ among tissues (e.g., muscle, tendon, bone, cartilage) and systems? In this paper, we discuss the "normal" tissue response to loading schema and demonstrate the complex interaction among training intensity, volume, and frequency, as well as the impact of these training variables on the recovery of specific tissues and systems. Although the response to training stress follows a predictable time course, the recovery of individual tissues to training load (defined herein as the readiness to receive a similar training stimulus without deleterious local and/or systemic effects) varies markedly, with as little as 30 min (e.g., cartilage reformation after walking and running) or 72 h or longer (e.g., eccentric exercise-induced muscle damage) required between loading sessions of similar magnitude. Hyperhydrated and reactive tendons that have undergone high stretch-shorten cycle activity benefit from a 48-h refractory period before receiving a similar training dose. In contrast, bone cells desensitize quickly to repetitive loading, with almost all mechanosensitivity lost after as few as 20 loading cycles. To optimize loading, an additional dose (≤ 60 loading cycles) of bone-centric exercise (e.g., plyometrics) can be performed following a 4-8 h refractory period. Low-stress (i.e., predominantly aerobic) activity can be repeated following a short (≤ 24 h) refractory period, while

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greater recovery is needed (≥ 72 h) between repeated doses of high stress (i.e., predominantly anaerobic) activity. The response of specific tissues and systems to training load is complex; at any time, it is possible that practitioners may be optimally loading one tissue or system while suboptimally loading another. The consideration of recovery timeframes of different tissues and systems allows practitioners to determine the "normal" response to load. Importantly, we encourage practitioners to interpret training within an athlete monitoring framework that considers external and internal load, athlete-reported responses, and objective markers, to contextualize load-response data.

Key Points

- We provide the typical timelines of recovery when exposing different tissues and systems to varying loading intensities, volumes, and frequencies.
- Muscle, tendon, and bone are highly stressed in some loading activities (e.g., sprinting). Given the different recovery timeframes, optimal loading may occur in one tissue while suboptimal loading may occur in another.
- We emphasize the importance of targeting the vulnerable tissue in rehabilitation programs and “training around” the injured tissue to maintain global load capacity and capacity of healthy tissues.

Hamstring Injuries, From the Clinic to the Field: A narrative review discussing exercise transfer. International Journal of Sports Physiology and Performance, Ahead of Print: 1-9, 2024.

Purpose: The optimal approach to hamstring training is heavily debated. Eccentric exercises reduce injury risk; however, it is argued that these exercises transfer poorly to improved hamstring function during sprinting. Some argue that other exercises, such as isometric exercises, result in better transfer to running gait and should be used when training to improve performance and reduce injury risk. Given the performance requirements of the hamstrings during the terminal swing phase, where they are exposed to high strain, exercises should aim to improve the torque production during this phase. This should improve the hamstrings’ ability to resist overlengthening consequently, improving performance and limiting strain injury. Most hamstring training studies fail to assess running kinematics postintervention.

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Of the limited evidence available, only eccentric exercises demonstrate changes in swing-phase kinematics following training. Studies of other exercise modalities investigate effects on markers of performance and injury risk but do not investigate changes in running kinematics.

Conclusions: Despite being inconsistent with principles of transfer, current evidence suggests that eccentric exercises result in transfer to swing-phase kinematics. Other exercise modalities may be effective, but the effect of these exercises on running kinematics is unknown.

Hamstrings are stretched more and faster during accelerative running compared to speed-matched constant-speed running. Medicine and Science in Sports and Exercise, Ahead of Print: 1-37, 2024.

Objective: Hamstring injuries are common in field-based sports and reinjury rates are high. Recent evidence suggests hamstring injuries often occur during accelerative running, but investigations of hamstring mechanics have primarily considered constant-speed running. Thus, our objective was to compare hamstring lengths and velocities between accelerative running and constant-speed running.

Methods: We recorded videos of 10 participants during 6 accelerative running trials and 6 constant-speed running trials. We used OpenCap to estimate body segment kinematics and a 3-dimensional musculoskeletal model to compute peak length and step-average lengthening velocity of the biceps femoris (long head) muscle-tendon unit. We compared running conditions using linear mixed models with running speed as the independent variable.

Results: At running speeds below 75% of top speed, accelerative running resulted in greater peak lengths than constant-speed running. For example, the peak hamstring muscle-tendon length when a person accelerated from running at only 50% of top speed was equivalent to running at a constant 88% of top speed. Lengthening velocities were greater during accelerative running at all running speeds. Differences in hip flexion kinematics drove the greater peak lengths and lengthening velocities observed in accelerative running.

Conclusions: Hamstrings are subjected to longer lengths and faster lengthening velocities in accelerative running than in constant-speed running. This provides a potential biomechanical perspective towards

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understanding the occurrence of hamstring injuries during acceleration. Our results suggest coaches and sports medicine staff should consider the accelerative nature of running in addition to running speed to quantify exposure to high-risk circumstances with long lengths and fast lengthening velocities of the hamstrings.

Is hip adductor or abductor strength in healthy athletes associated with future groin pain? A systematic review and meta-analysis. British Journal of Sports Medicine, Ahead of Print: 1-9, 2024.

Objective: To systematically review the association between hip adduction and abduction strength, and adduction-to-abduction strength ratio in healthy athletes with the occurrence of future groin pain and time-loss groin injuries.

Design: Systematic review with meta-analysis.

Data sources: PubMed, Web of Science, SPORTDiscus, PEDro, Embase, and Scopus.

Eligibility criteria: Studies included had to (1) measure hip adductor or abductor strength in healthy athletes, (2) conduct a follow-up period to assess the occurrence of groin pain, (3) present strength data separately for participants who remained uninjured and those who suffered an injury and (4) use a prospective design. Risk of bias was assessed according to the Quality in Prognosis Studies tool. The certainty in the evidence appraised was measured using the Grading of Recommendations Assessment Development and Evaluation approach.

Results: Thirteen prospective cohort studies met the inclusion criteria. Overall study risk of bias was rated as low. Players who remained uninjured had stronger adduction strength compared with players who subsequently suffered groin pain (standardised mean differences with 95% CIs (SMD=-0.5, 95% CI -0.92 to -0.09)) and time-loss groin pain (SMD=-0.68, 95% CI -1.31 to -0.06). Trivial effects were observed for abduction strength (groin pain SMD=0.03, 95% CI -0.11 to 0.17; time-loss SMD=-0.07, 95% CI -0.25 to 0.11) and adduction-to-abduction strength ratio (groin pain SMD=-0.02, 95% CI -0.55 to 0.51; time-loss SMD=-0.11, 95% CI -1.11 to 0.89). Age and diagnostic criteria were not significant moderators of the relationship between adductor strength and groin pain (p=0.72 and p=0.12).

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Conclusion: There is a moderate effect of hip adductor strength on the occurrence of groin pain, while there is no relationship between either abductor strength or the ratio of adductor-to-abductor strength with the occurrence of groin pain.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Groin problems are among the most common injuries in multidirectional sports.

⇒ Hip strength plays a vital role in the prevention and rehabilitation of groin pain.

WHAT THIS STUDY ADDS

⇒ Hip adductor weakness is a potential risk factor for groin pain.

⇒ Neither abductor strength nor the adductor-to-abductor strength ratio was significantly related to the occurrence of subsequent groin pain.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ These findings suggest that athlete screening should incorporate adduction strength testing to identify individuals with lower strength values and implement injury prevention strategies.

Multi-modal approach to mitigating hamstring injuries in Division 1 college football athletes. Encyclopedia, 4: 1482-1495, 2024.

Definition - Hamstring injuries (HSIs) are prevalent in sports that involve changes in direction, kicking, and sprinting. These injuries are a major cause of time lost from competition, practice, and training, as well as increased healthcare costs. In a Division I collegiate football program, the authors implemented a multifactorial approach that included repeated performance assessments, detailed data analysis, and a flexible strength and conditioning regimen. Over a three-year period, this resulted in no game time loss due to HSI. This model can be adapted and implemented across sports settings.

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NFL Quarterback Development – The role of competing quarterbacks, rookie playing time, and team quality. The American Economist, Ahead of Print: 1-16, 2024.

Abstract - Using data from 1994 to 2017, we consider characteristics that affect quarterback (QB) development in the National Football League. Our analysis leads to three key findings. Firstly, we show that a young QB's career is worsened when he plays alongside a Pro Bowl QB. Secondly, we find that QB's career outcomes are better when they are drafted by a bad team. Finally, we investigate the role of rookie playing time and find evidence that playing as a rookie reduces career outcomes. In all three cases, the effects are largest in the latter portion of a QB's career.

Optimizing acute static stretching protocols: Balancing range of motion and strength performance. Current Sports Medicine Reports, 23(11): 363-365, 2024.

Introduction: In the warm-up routine before competitions or training sessions, many authoritative sports science organizations advocate for the use of dynamic stretching rather than acute static stretching. This recommendation stems from evidence indicating that acute static stretching can reduce athletic performance, particularly in areas related to strength and explosive power. For sports that do not require a large range of motion (ROM) in joints, dynamic stretching is preferable. However, when a significant ROM is necessary, acute static stretching remains essential despite its potential to diminish athletic performance. To support athletes who must incorporate acute static stretching into their warm-up routines, this paper explores the factors contributing to the negative impacts of acute static stretching and outlines protocols that can preserve ROM benefits while mitigating adverse effects. When developing stretching protocols, it is crucial to consider two fundamental aspects: duration and intensity. In practical warm-up routines, athletes typically engage in sport-specific dynamic exercises and other pre-competition preparations after stretching. Consequently, a warm-up consisting solely of acute static stretching is rare, and there is usually a time interval between stretching and the subsequent sports activity (i.e., training sessions or competitions). Therefore, designing effective acute static stretching protocols requires consideration of several factors: the intensity and duration of stretching, its combination with dynamic exercises, and the interval between warming up and engaging in sports

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activities. These factors collectively influence the overall effect of acute static stretching on athletic performance.

Conclusion: To minimize the potential negative impact of acute static stretching on subsequent athletic performance, we recommend considering the duration of stretching, intensity, dynamic activities before and after stretching, and the interval between stretching and athletic activities when designing acute static stretching protocols. Specifically, a protocol where the stretching time is 60 s, divided into at least two sets, with intensity controlled at 100% POD, and includes targeted dynamic exercises before and after stretching, can enhance ROM for up to 30 min without negatively affecting athletic performance, and may even improve it. Notably, if the stretching duration needs to be extended to 120 s, the dynamic exercises following stretching may need to be adjusted to activities that can induce post-activation performance enhancement to ensure that acute static stretching does not have adverse effects.

Shoulder injury patterns and trends in the National Football League: An epidemiological analysis. Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-8, 2024.

Background: Athletic movements in American football render the shoulder vulnerable to injury. The aim of this study was to explore the epidemiology of shoulder injuries in the National Football League (NFL).

Methods: Injury data from the 2019-2022 seasons were extracted from official NFL website and secondary resources. Injury parameters included type, severity, week, position and other demographic characteristics. Injury rates were calculated per 1000 athletic exposures (AE) and statistical analyses

were used to determine whether any significant differences exist in injury patterns across different positions and types.

Results: A total of 6732 injuries were reported during our study period, of which 452(6.7%) were attributed to the shoulder, leading to an injury rate of 4.5 per1000 AE. Linebackers, cornerbacks, wide receivers, and safety positions had significantly higher numbers of shoulder injuries than other defensive and offensive positions. Most injuries were mild (66%, $P<0.001$). Muscular (40.9%) and ligamentous (37.4%) injuries were the most common injury types. Fractures and ligamentous injuries were

significantly more severe than muscular and other injuries (P<0.05). Even though week 17 had the highest mean number of shoulder injuries at 10.75/season, there was no apparent shoulder injury distribution trend across the football season.

Conclusions: Shoulder injuries are common in American football. Introducing a deidentified injury registry that provides detailed injury information in the NFL may help develop better prevention strategies and protocols.

Single nucleotide polymorphisms and tendon/ligament injuries in athletes: A systematic review and meta-analysis. International Journal of Sports Medicine, Ahead of Print: 1-19, 2024.

Abstract - This systematic review and meta-analysis aimed to identify the association between genetic polymorphisms and tendon and ligament injuries in adolescent and adult athletes of multiple competition sports. The PubMed, Web of Science, EBSCO, Cochrane Library, and MEDLINE databases were searched until July 7, 2023. Eligible articles included genetic studies on tendon and ligament injuries and comparisons between injured and non-injured athletes. This review included 31 articles, comprising 1,687 injury cases and 2,227 controls, from a meta-analysis of 12 articles. We identified 144 candidate gene polymorphisms (only single nucleotide polymorphisms were identified). The meta-analyses included vascular endothelial growth factor A (VEGFA) rs699947, collagen type I alpha 1 rs1800012, collagen type V alpha 1 rs12722, and matrix metalloproteinase 3 rs679620. The VEGFA rs699947 polymorphism showed a lower risk of injuries in athletes with the C allele ([C vs. A]: OR=0.80, 95% CI: 0.65-0.98, I 2 =3.82%, p=0.03). The risk of these injuries were not affected by other polymorphisms. In conclusion, the VEGFA rs699947 polymorphism is associated with the risk of tendon and ligament injuries in athletes. This study provides insights into genetic variations that contribute to our understanding of the risk factors for such injuries in athletes.

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Spatiotemporal kinematics during top speed sprinting in male intercollegiate track and field and team sport athletes. Sports Biomechanics, Ahead of Print: 1-15, 2024.

Abstract - We investigated spatiotemporal kinematics during top speed sprinting and biomechanical running strategies in 98 male intercollegiate athletes from a range of athletic backgrounds in track and field (TF, n = 28) and team sports (TS, n = 70). Participants completed 40 m running trials with sagittal plane motion analyses of high-speed video captured from 30 m to 40 m. Across the entire sample, measures of contact time, step rate, step length, flight length and duty factor (ratio of contact duration to stride duration) were meaningfully correlated with top speed ($p < 0.05$, $0.51 \leq |r \text{ or } \rho| \leq 0.78$). Flight time and contact length were weakly correlated with top speed ($p < 0.05$, $0.27 \leq |r \text{ or } \rho| \leq 0.34$). When comparing sub-groups of Slow TF (n = 14) and Fast TS athletes (n = 22) with similar top speeds (~9.3 m/s), Fast TS athletes clearly demonstrated a more ground-based strategy, with longer ground contact times and contact lengths, shorter flight times and flight lengths, and larger duty factors. Therefore, the results of this study suggest that existing technical models and normative metrics based on data from TF athletes could require modification when evaluating and coaching sprinting performance with TS athletes.

Sport-related injury and performance enhancing substance use by young athletes. Sports Health, Ahead of Print: 1-15, 2024.

Background: Legal performance enhancing substance (PES) use is increasing in young athletes and may lead to banned PES use. This study examines legal PES use and consideration of banned PES use in young athletes with a season-ending injury (SEI) compared to young athletes with non-SEI/no previous injury.

Hypothesis: Young athletes sustaining SEI or concussions have increased odds of reporting legal PES use and consideration of banned PES use compared with non-SEI or no injury.

Study design: Cross-sectional study.

Level of evidence: Level 4.

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Methods: Cross-sectional study from 2013 to 2020 of athletes aged 6 to 25 years.

Results: No association was found between reporting legal PES use or consideration of banned PES use and young athletes with SEI or concussion compared with no injury/non-SEI. 14% of athletes reported legal PES use and 3% reported consideration of banned PES use. In adjusted analysis, athletes who were male (odds ratio [OR], 1.38; 95% CI, 1.16-1.73; P = 0.03), have history of depression (OR, 3.01; 95% CI, 1.26-7.18; P = 0.01), weight train (OR, 1.66; 95% CI, 1.04-2.65; P = 0.03), and believe that athletic ability is influenced by weight (OR, 1.75; 95% CI, 1.08-2.83; P = 0.02) had increased odds of reporting legal PES use. Older (OR, 1.18; 95% CI, 1.03-1.36; P = 0.02) and male (OR, 1.57; 95% CI, 1.02-1.81; P < 0.04) athletes had increased odds of consideration of banned PES use.

Conclusion: This study found no increased odds of reported PES use in young athletes with sports-related injury or concussion regardless of injury type or severity. Factors associated with PES use were male sex, age, history of depression, weight training, and belief that weight affects athletic performance.

Clinical relevance: Risk factors associated with PES use in young athletes are essentially unknown. This study can inform pertinent clinical care, education, and policy implementation.

Summary of the 2024 Professionals in Nutrition for Exercise and Sport "10 Questions/10 Experts" Session – Hot Topics for the Paris Olympic Games. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-8, 2024.

Overview - This article provides a recap of the 10 Questions/10 Experts session at the 2024 American College of Sports Medicine Annual Meeting. Each of the speakers considered the validity of common "myths," while providing evidence-based opinions to support, or, bust, myths addressing the following questions: (1) Would 100 g/hr of carbohydrate be advisable for the Olympic Cycling Road race? (2) Is there an advantage in the marathon of ingesting bicarbonate as a hydrogel product? (3) Can genotyping be used to individualize caffeine supplementation in football? (4) Should low fluid consumers drink more to improve 1,500-m track performance? (5) Do urinary markers of dehydration predict poor basketball performance? (6) Do placebo effects influence 10-km track performance? (7) Should combat athletes

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make weight using glucagon-like peptide-1 receptor agonists? (8) Would crushed ice ingestion help tennis umpires make better decisions in the heat? (9) Are collagen supplements useful to reduce tendon and ligament injuries in volleyball? and (10) Should female athletes plan their training and diet according to their menstrual cycle? This article describes the content of each of the presentations including the most important outcomes and conclusions drawn by the presenters.

The impact of daylight exposure on injured athletes: Implications for rehabilitation. International Journal of Strength and Conditioning, 4(1): 1-15, 2024.

Abstract - During injury rehabilitation and return to competition, various factors impact outcomes, such as psychological parameters, sleep quality, and vitamin D levels. Studies have demonstrated that both natural and artificial daytime lighting can enhance these variables, as well as human health and performance. However, research into therapeutic methods is predominantly restricted to physical or psychological interventions that are costly and time-consuming, rather than integrated solutions. Research on the impact of daylight on injury rehabilitation among athletes is limited. Objective evaluation is necessary to establish effective recovery protocols, providing the rationale for this review. Therefore, this study analyzes existing research and theoretical explanations regarding the effects of daylight on athletes in the rehabilitation process. The results suggest that the underlying mechanism can be organized into sun induced vitamin D synthesis, cognitive-behavioral processes, the circadian mechanism, and the visual processing of light stimuli. For practical implementation, we propose a cost-effective and time-saving artificial daylight intervention that can serve as an accompanying therapy for injuries. This intervention could be integrated into therapy facilities and sports clubs, for both competitive and recreational sports.

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The increasing medicolegal cost in the care of high-level athletes. Journal of Shoulder and Elbow Surgery. Ahead of Print: 1-18, 2024.

Background: Medical malpractice represents a significant economic cost in health care. Increasingly large damage claims by professional athletes against physicians have raised concerns about the medicolegal challenges in caring for high-level athletes.

Methods: An online proprietary legal research database was queried for lawsuits related to malpractice in the care of professional and amateur athletes from 1992 to 2023. Demographics of plaintiffs and defendants, details of lawsuit filings, and damages claimed were recorded for all cases, settlements, and jury verdicts. Descriptive statistics, linear regression, as well as univariate analysis of demographic factors and damage claims in inflation-adjusted 2023 dollars was performed.

Results: Eighty-eight lawsuits were recorded from 1992 to 2023. The sum of indemnities exceeded 186 million in 2023 inflation-adjusted United States Dollars. Football players were the most commonly represented athletes (n = 26) and represented 49% of total financial awards. Fourteen cases (16%) involved treatment of the upper extremity. Professional and collegiate level of play was associated with higher damages in favor of plaintiffs when compared to other levels of play. No other demographic was associated with higher financial awards. Linear regression showed a significant positive trend with an increasing inflation-adjusted compensation for plaintiff verdicts in the last 30 years.

Conclusions: There is an increasing medicolegal financial risk associated with the care of athletes. This is higher in collegiate and professional levels of play. As physicians, insurers and institutions adjust to these financial risks, care must be taken to avoid ramifications on the availability and quality of care provided to athletes. Shoulder and elbow surgeons may consider additional preoperative counseling, legal waiver forms regarding malpractice claims, and advocacy for medical malpractice reforms in the care of athletes with high earning potential to mitigate these increasing financial risks.

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The injury mechanism of Achilles tendon rupture in professional athletes – A video analysis study. Journal of the American Podiatric Medical Association, 114(3): 1-7, 2024.

Background: In professional athletes, Achilles tendon ruptures are devastating injuries, often resulting in season loss or decreased return to sports level. We aimed to perform a comprehensive video analysis to describe the mechanisms of Achilles tendon rupture as well as body posture in professional athletes. We hypothesized that Achilles tendon ruptures in professional athletes develop with a specific injury mechanism and that body posture at the time of injury varies by sport.

Methods: Of 114 identified Achilles tendon ruptures that occurred in professional athletes between 1970 and 2020, 42 with adequate video data were analyzed for injury mechanism, body posture, and player and sport characteristics by three independent reviewers.

Results: Mean ± SD age of the 42 athletes (41 men and 1 woman) was 28.4 ± 4.26 years. Eighteen athletes played basketball (42.9%), 14 football (33.3%), five soccer (11.9%), three baseball (7.1%), and two rugby (4.8%). Thirty-five patients (83.3%) had noncontact injuries. The most frequent months of injury were January (16.7%) and October (14.3%). During Achilles tendon rupture, the ankle was in dorsiflexion, the body was bent forward, the knee and hip were in extension, and the foot was in a neutral position. Most injuries occurred during take-off/acceleration (40.5%) or stop and turn (38.5%) maneuvers.

Conclusions: In professional athletes, the Achilles tendon most often ruptures during take-off/acceleration. The most common position during rupture is the trunk in flexion, the knee and hip in extension, and the ankle in dorsiflexion. This information can guide professional athletes in terms of physical therapy techniques, including neuromuscular training, proprioception, and balance training specific to preventing Achilles tendon rupture.

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To challenge or not to challenge? Significance, 21(1): 14-16, 2024.

Overview - In any sport, challenging a referee's decision is risky. The challenge process in American football is particularly chaotic, and coaches have just precious seconds in which to decide whether to throw the red challenge flag and gamble their team's fortunes. Can statistics help them?

Conclusion - By comparing a coach's red flag challenge to the casino game of craps, there is a simple model that helps determine the parameters for making a challenge and that provides the coach and his staff with a straightforward strategy for when to throw the red flag. If the coach's estimates are good, the strategy provides a gamble with positive expectation. This not only provides the coaches with a simple guideline for making a red flag challenge, but also indirectly increases the fans' enjoyment of watching a game by knowing that the coaches have a tool to help improve the accuracy of the outcome.

To jog or to stride: A comparison between two standardized running tests to monitor neuromuscular status in team sport athletes. Journal of Sports Sciences, Ahead of Print: 1-8, 2024.

Abstract - This study investigated: 1) monitoring post-match neuromuscular fatigue (NMF) status in rugby union players using two submaximal running tests (SRT); and 2) the sensitivity of each SRT to locomotor variables obtained during match-play. Twenty-three male rugby players (age: 21.0 ± 1.3 years; height: 185.2 ± 6.1 cm; body mass; 97.3 ± 10.3 kg) were monitored across one season (n = 71 player-match and 159 fatigue-players-testing observations). Two different SRTs (SRT-jog [5-minute shuttle run protocol] and SRT-stride [a repeat stride effort protocol]) were used to characterize post-match NMF, with measures taken two days prior to match day (baseline), on MD + 1, and MD + 2. Linear mixed models (±90% CIs) were used to explore differences between measures and match-play locomotor variables. SRT-jog presented a meaningful increase following MD + 1 (ES: 0.63 [0.37 to 0.89]). SRT-Stride showed small increases at MD + 1 (ES: 0.24 [0.02 to 0.49]) and MD + 2 (ES: 0.33 [0.07 to 0.59]) suggesting a potential impairment in running mechanics. SRT-stride was significantly associated with total distance (ES: 0.41 [−0.01 to 0.83]) and collisions (ES: 0.58 [0.18 to 0.99]). While both tests presented small to moderate changes post-match, only the SRT-stride was related to match locomotor

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variables. Therefore, the SRTstride may be a more sensitive monitoring tool for monitoring NMF in rugby union players.

Towards Achilles Tendon Injury Prevention in Athletes with Structural MRI Biomarkers: A Machine Learning Approach. Sports Medicine – Open, 10: 118, 2024.

Background: Recent advancements in artificial intelligence have proven their effectiveness in orthopaedic settings, especially in tasks like medical image analysis. This study compares human musculoskeletal radiologists to artificial intelligence in a novel, detailed, short, and cost-effective examination of Achilles tendon magnetic resonance images to uncover potential disparities in their reasoning approaches. Aiming to identify relationships between the structured assessment of the Achilles tendon and its function that could support injury prevention. We examined 72 athletes to investigate the link between Achilles tendon structure, as visualized in magnetic resonance images using a precise T2*-weighted gradient echo sequence with very short echo times, and its functional attributes. The acquired data were analyzed using advanced artificial intelligence techniques and reviewed by radiologists. Additionally, we conducted statistical assessments to explore relationships with functional studies in four meaningful groups: dynamic strength, range of motion, muscle torque and stabilography.

Results: The results show notable linear or non-linear relationships between functional indicators and structural alterations (maximal obtained Spearman correlation coefficients ranged from 0.3 to 0.36 for radiological assessment and from 0.33 to 0.49 for artificial intelligence assessment, while maximal normalized mutual information ranged from 0.52 to 0.57 for radiological assessment and from 0.42 to 0.6 for artificial intelligence assessment). Moreover, when artificial intelligence-based magnetic resonance assessment was utilized as an input, the associations consistently proved more robust, or the count of significant relationships surpassed that derived from radiological assessment. Ultimately, utilizing only structural parameters as inputs enabled us to explain up to 59% of the variance within specific functional groups.

Conclusions: This analysis revealed that structural parameters influence four key functional aspects related to the Achilles tendon. Furthermore, we found that relying solely on subjective radiologist

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opinions limited our ability to reason effectively, in contrast to the structured artificial intelligence assessment.

Study design: Cross-sectional studies.

Validity of four consumer-grade optical heart rate sensors for assessing volume and intensity distribution of physical activity. Scandinavian Journal of Medicine & Science in Sports, 34: e14756, 2024.

Abstract - Consumer-grade optical heart rate (HR) sensors emerged as promising tools to monitor volume and intensity of physical activity (PA). However, no validation study of optical HR sensors included recent comprehensive validation recommendations, required for facilitating usage for medical purposes. Validity of HR data measured with four consumer-grade optical sensors, the wrist-worn Garmin Venu 2S and Polar Vantage M2, and the upper arm-worn Polar Verity Sense and Scosche Rhythm24 were assessed in 32 participants over 24 h including various laboratory-based and free-living activities. Furthermore, validity of time at moderate or vigorous PA intensity zones was analyzed. A medical-grade ECG served as a reference. Reliability was assessed by analyzing data of laboratory-based activities during two visits. Across activities, the mean absolute percentage error ranged from 2.2% to 4.7% and intraclass correlation coefficients ranged from 0.91 to 0.98, indicating high validity for all optical sensors. All sensors validly detected time spent at moderate or vigorous intensities (mean error < 10%), except Polar M2 for moderate PA (mean error 12.8%) and Garmin V2 for vigorous PA (mean error -15.9%). Sensor day-to-day reliability was high, indicated by a mean absolute error of < 5 beats/min. Upper arm-worn sensors consistently outperformed wrist-worn sensors, particularly in activities involving increased arm movement and at higher intensities. Our findings identified an overall high validity of these four consumer-grade optical HR sensors, also for assessing time spent at higher PA intensities. However, differences were observed between sensors regarding activity subtypes and intensity levels. These data support informed decisions when selecting optical sensors for PA monitoring and intervention. Trial Registration: The study was performed in compliance with the Declaration of Helsinki and its current amendments and was prospectively registered at ClinicalTrials.gov (ID NCT05525000).

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